

Tsukorok S3 v4 firmware 5.5.12

Detection methods, alarm decision logic, evidence ranking and TDOA suitability

Source: static analysis of tsukor_s3v4_5.5.12_en.bin (Ghidra 11.4.2, Xtensa) and the AMICCOM A5133 datasheet v0.7. Full addresses and raw constants are in research/briefs/tsukorok-firmware-5.5.12-rf-extraction.md. Values marked CONFIRMED were read from bytes or disassembly; INFERRED means reasoned from behaviour. "TDAO" in the request is read as TDOA (time difference of arrival localisation); if a different system was meant, section 5 must be re-read.

1. What the device is

An ESP32-S3 handheld with three receivers driven by RadioLib and a bit-banged driver: an SX1280 for 2.4 GHz, an SX1276/SX1278 for 860-1020 MHz, and an AMICCOM A5133 for 5.8 GHz. No protocol is decoded anywhere.

Every alarm is produced either from RSSI sweeps (energy morphology) or from the first 32-bit word received after an FSK preamble ("syncword"). In AERIX RF terms every output is stage-1 (RF candidate/morphology) or stage-2 (probabilistic classification) evidence.

One caveat for anyone using it as a reference receiver: the 5.8 GHz driver issues the A5133 TX-mode strobe (0xD8) to leave RX before each RSSI read and holds TX mode for 10 ms at every scan start and before calibration. Whether that radiates has not been measured, but the device should not be assumed silent at 5.8 GHz.

2. Detection methods and the yes/no alarm rule

Each row gives the radio, what is measured, the exact decision, and the label the device shows. Thresholds are the firmware defaults.

#	Method (radio)	What is measured	Alarm = YES when	Label / type
A1	Sub-GHz FSK static syncword (SX127x)	FSK RX at 57.6 / 80 / 15.235 kbps, 100 kHz steps over 860-885 / 895-928 / 970-1020 MHz. First 32-bit word after preamble, normalised, looked up in a 9-entry table.	Word (or its shifted/normalised form) matches the table AND corrected RSSI $\geq$ -94 dBm (rssi_threshold 188 / 2). One packet is enough.	Or (Orlan class), Za (Zala), ZL (ZalaLancet), EI (ELRS-like)
A2	Sub-GHz syncword-count (learned)	Same packets; every new word is stored with frequency, bitrate, time (entries expire after 20 min).	Same word seen again: count $\geq$ 2 on 865-872 or 902-928 MHz with a second such word in the band $\rightarrow$ ZL; same word on both bands at 15.235 kbps $\rightarrow$ Lc; at 84.6-85.6 kbps in 860-885 or 902-922 MHz $\rightarrow$?FPV; at 15.235 kbps above 970 MHz $\rightarrow$ Sc. Plus the RSSI gate.	ZL, Lc, ?FPV, Sc; otherwise "*" (unknown FSK, type 1)
A3	"cryptoorlan" (learned, 55.0-58.0 kbps only)	Words that differ on every packet at $\sim$ 57.6 kbps on one frequency.	More than 3 distinct words stored AND more than 3 of them seen exactly once (i.e. $\geq$ 4 different words at one frequency), packet RSSI arg > 30.	X3 (type 10)
B1	Sub-GHz LoRa / ELRS mode 1 (SX127x CAD)	Per 125 kHz channel: RSSI gate, then LoRa channel-activity detection at SF6-9 with 500 kHz BW for 50 ms; if two CAD hits at one SF, receive 8 bytes and read SNR.	At the end of the range: max RSSI $\geq$ -199 (always) AND 6·good + 3·bad + timeouts $\geq$ 6 AND the dominant SF was hit more than {SF6:5, SF7:8, SF8:5, SF9:4} times.	"Lora detected" match callback (ELRS-like)
B2	Sub-GHz LoRa mode 2	Same, single SF (sf_first = 9), 2-3 passes.	6·good + 3·bad $\geq$ 6 AND (good $\geq$ 1 OR bad $\geq$ 2).	same
C1	2.4 GHz DJI spectral score A (SX1280)	8 sweeps of 256 bins (390.625 kHz) over 2400-2500 MHz. Segments above median+5 dB, features: width, mean-median, flatness, persistence over sweeps.	Score = 0.55·s1 + 0.45·s2 > 40 (of 100). Width ideal 39 bins ($\sim$ 15 MHz), accepted 26-52 bins; persistence in $\geq$ 3 of 8 sweeps.	DJI (type 13) or DJIe (type 18, width > 80 bins)
C2	2.4 GHz DJI spectral score B	Same spectrum; segments 18-60 bins wide, gap merge 3, persistence $\geq$ 3 sweeps with $\geq$ 2 bins, $\geq$ 85 % of cells above threshold, $\leq$ 40 % always-on bins, centre-edge margin 5 dB.	Score > 3.14. dji_alg = 2 needs A AND B, dji_alg = 3 needs A OR B.	DJI / DJIe

#	Method (radio)	What is measured	Alarm = YES when	Label / type
C3	2.4 GHz DJI candidate + time-domain (dji_alg = 1, default)	Candidates: bins above 40th-percentile+6 dB in ≥ 2 sweeps, width 20-102 bins. For each candidate 8192 RSSI samples on two channels (start+w/3, end-w/3).	Both traces classified "periodic bursts": duty 0.3-0.9, max gap ≤ 750 samples, < 3 short bursts (32-75 samples), ≥ 5 long bursts and gaps (≥ 64 samples), modal burst/gap length (20-sample buckets) repeats $\geq 5/\geq 3$ times, top buckets cover $\geq 50\%$.	DJI / DJIe
C4	2.4 GHz "paranoid" sweep (dji_alg = 4)	250 channels x 400 kHz, 40 μ s settle, threshold dji_rssi_thrsh -95 dBm; 16-sample sliding window.	≥ 8 of the last 16 channels above threshold (≥ 3.2 MHz) AND fewer than 125 of 250 channels above threshold, accumulated over sweeps: +1/-1 per sweep, clamp 0-4, alarm when accumulator > 1 .	DJI
C5	2.4 GHz Zala video sweep (detect_zl_la = 2)	500 channels x 400 kHz from 2200 MHz, 60 μ s settle, threshold -94 dBm.	≥ 7 of 16 consecutive channels above threshold AND fewer than 250 above; accumulator clamp 0-3, alarm when > 1 .	ZalaV (type 2)
C6	"Skydio" metric (skydio_alg 1-4)	Histogram of the 3-dB-quantised RSSI levels above -90 dBm in the 8x256 spectrum.	Count of the most frequent level > 70 gives 80, > 50 gives 60, > 30 gives 40; alarm if that value $\geq \{70,60,50,40\}[\text{skydio_alg}]$.	Skydio (type 5)
D1	5.8 GHz A5133 paranoid sweep (dji_alg = 4)	175 channels x 1 MHz, 5725-5899 MHz, RSSI = ((ADC-RL)/(RH-RL)) $\cdot 12 - 83$ dBm, threshold -95 dBm.	Same 16-window rule (≥ 8 of 16), guard < 87 channels, accumulator 0-4, alarm > 1 ; OR immediately if the 16-sample window shows a rise-peak-fall shape (returns "dji 5.8").	FPV5 / DJI 5.8 (type 3)
D2	5.8 GHz A5133 spectral (other dji_alg)	One 175-channel sweep; thresholds from -65 dBm down to -99 in 5 dB steps.	A segment 16-69 MHz wide (gap tolerance 5) whose edges stay within 5 MHz of the segment found at the previous, higher threshold (steep edges); needs two consecutive positive sweeps.	FPV5 (type 3)
E1	Track mode (SX1280 or A5133)	$f \pm 10$ MHz around a chosen frequency, 400 kHz (2.4 GHz) or 1 MHz (5.8 GHz) steps, two passes.	More than 10 samples above threshold.	Track f=... r=...
E2	RSSI scan / unknown drone (SX127x)	RSSI sweep of the configured range at 100 kbps FSK, +15 dB correction if $> 30\%$ of channels read below -112 dBm.	Peak $> -\text{rssi_scn_tresh}$ (-60 dBm).	Unknown

Common gate for every sub-GHz alarm: a packet only counts if RSSI (after the +15 dB noise correction) is at least $-(\text{rssi_threshold}/2) = -94$ dBm; below that it is logged as "below threshold" and discarded. All 2.4/5.8 GHz alarms are rate-limited to one per type per 3 s.

3. Pseudocode of the two decision cores

Sub-GHz FSK packet (IRAM FSK_MatchOnPacket):

```
rssi = read_rssi(); if noise_flag: rssi += 15
hits = [static_table(word), sync_count(word, f, br), cryptoorlan(word, f, br, rssi)]
if hits empty: log "candidate f, br" # stage-1 only, no alarm
elif rssi < -(rssi_threshold >> 1): log "below threshold"
else: alarm(label(hits, br), f, rssi) # one packet suffices
```

2.4 / 5.8 GHz tick (IRAM TwoFourGScanner_Tick, at most every 5 s):

```
if dji_alg == 4: det = paranoid_sweep()
else: spec = 8 sweeps x 256 bins
A = scoreA(spec) > 40 ; B = scoreB(spec) > 3.14
dji_alg 2: det = A and B ; 3: det = A or B
dji_alg 1: det = any(candidate c: burst_classify(td(c.start+w/3)) and
burst_classify(td(c.end-w/3)))
if not det and skydio_alg: det = noise_level(spec)  $\geq$  table[skydio_alg]
if not det and detect_zl_la == 2: zala = zala_sweep()
if not zala and a5133: r = a5133_sweep(); five8 = (r == 2); det |= (r != 0)
emit(type 3 if five8 else 2 if zala else 1/5) unless same type within 3000 ms
```

4. Ranking by evidence value

Ranked by how far the rule is from "there is energy here" towards a system-specific fingerprint, and by the false-alarm exposure of the rule. This is an assessment (INFERRED), the thresholds are CONFIRMED.

Rank	Method	AERIX RF evidence level	Why
1	A1 static syncword + bitrate/band	Level 3 candidate (protocol-specific evidence) if the nine words are genuine; until verified, level 2	A 32-bit word after a preamble at a specific bitrate in a specific band is a real fingerprint. Weakness: the words come from the vendor, are not validated in the image, and a single packet fires the alarm.
2	A2 syncword-count with band rules	Level 2	Repetition of the same unknown word on the same channel is strong evidence of a real digital link; the band/bitrate rules add plausibility but the "which drone" labels are heuristic.
3	B1 LoRa CAD + SNR (ELRS-like)	Level 2	CAD at SF6-9/500 kHz plus a positive-SNR packet is a good LoRa-presence test; it does not distinguish ELRS from other LoRa users.
4	A3 cryptoorlan	Level 2, high false-alarm exposure	Four different words at ~57.6 kbps is consistent with an encrypted/whitened link but also with any noisy FSK demodulation; keep as supporting evidence only.
5	C3 candidate + time-domain bursts	Level 1-2	Combines width with burst periodicity; the best of the 2.4 GHz rules, but the periodicity test is coarse (20-sample buckets, no timing reference).
6	C1 / C2 spectral scores	Level 1	Width, flatness and persistence of a 7-23 MHz block. Wi-Fi, video links and other OFDM users share this morphology.
7	D2 5.8 GHz spectral (steep edges)	Level 1	A steep-edged 16-69 MHz block in 5.8 GHz is typical of analog/digital FPV video; not specific.
8	C4 / C5 / D1 paranoid sweeps, E1 track	Level 1	Pure occupancy counting in a 16-channel window. The broadband guards only reject jamming-like saturation.
9	C6 "Skydio" metric, E2 RSSI scan	Below level 1	A histogram of RSSI levels or a peak above -60 dBm says nothing about the source.

5. Usability for TDOA

TDOA needs the same emission time-stamped at three or more receivers with a common clock, at a precision of tens of nanoseconds (300 m of range difference per microsecond). Judged against that:

- **None of the device's own detection outputs are TDOA-usable.** They are RSSI sweeps with 40-300 μ s settle times and 5 s tick periods, or packet events time-stamped by *millis()* on an unsynchronised ESP32. There is no IQ, no sample clock, no GPS-disciplined time.
- **The signals it exploits are TDOA-usable with SDRs.** Sub-GHz FSK packets with a fixed preamble and a repeating 32-bit word (A1/A2) give a correlation peak that can be time-stamped at the sample level; ELRS LoRa packets (B1) likewise via chirp correlation; DJI OcuSync bursts (C-methods) are wide, short and periodic and are the classic TDOA target. What is needed is coherent, GPS-disciplined receivers such as the ANTSDR E200 with a common reference, not this device.
- **What can be reused:** the frequency plans, bitrates and syncwords as *where to correlate*; the burst-periodicity idea (C3) as a gate before spending TDOA correlation effort; the DJI width window as a candidate filter.
- **RSSI-only alternative:** the device's outputs could feed RSSI-based ranging or direction-by-comparison across several units, which is not TDOA and is far less accurate (± 6 dB RSSI on the A5133 alone).

Summary: use the firmware as a catalogue of heuristics and target signatures; use the E200/HackRF chain for anything that must be timed.